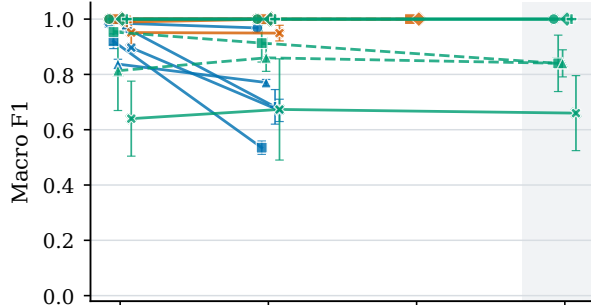


Scaling by tier across six models

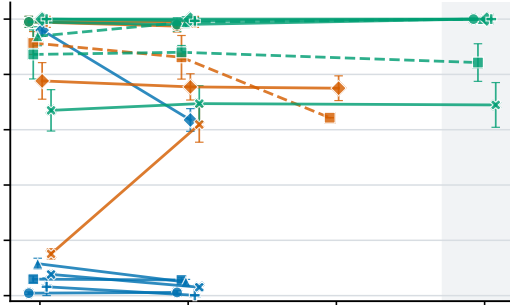
LLM CodeAct RLM

● Qwen 3.5 ■ *DeepSeek V4 Flash ▲ *GLM 5.2 ◆ Gemini 3.7 Flash + GPT-5 mini ✖ Claude Haiku 4.5

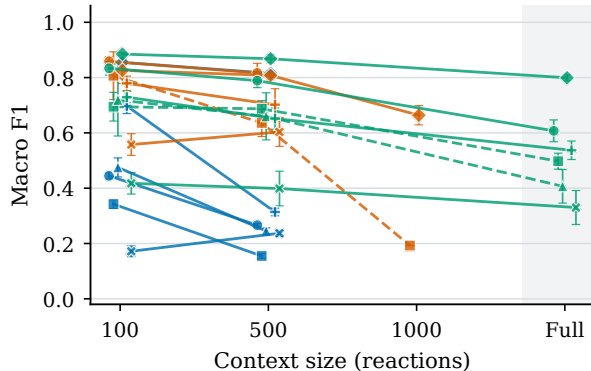
(a) Tier 1: Structural lookup ($n=10$)



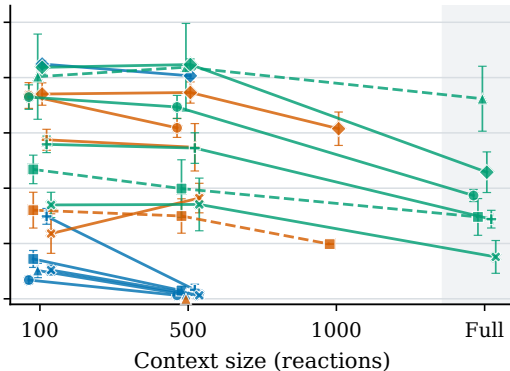
(b) Tier 2: Property aggregation ($n=20$)



(c) Tier 3: Reaction-level reasoning ($n=35$)



(d) Tier 4: Relational reasoning ($n=35$)



Terminal failed jobs score zero; unresolved jobs are excluded. Small offsets improve visibility; dashed curves and * labels are provisional.